

DECODED EXPLAINED

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Problem With Ethanol Blending Isn't Sugar — It's Reliance on Grains

Problem with ethanol blending isn't sugar — it's reliance on grains; the way forward



HARISH DAMODARAN

INDIA'S ETHANOL blended petrol (EBP) programme was initiated primarily to benefit the sugar industry. For mills confronting stagnant sugar consumption demand amid rising production, ethanol created an additional revenue stream to enable them to make timely payments to cane farmers.

Until 2017-18, all ethanol supplied to oil marketing companies (OMC) came from "C-heavy" molasses — a thick, dark brown final byproduct of cane-juice processing, containing sucrose that mills cannot further economically recover as sugar after three rounds of boiling and crystallisation. From the 2018-19 supply year, mills started producing ethanol from the intermediate-stage "B-heavy" molasses and also directly from whole sugarcane juice syrup. With less sucrose getting crystallised and recovered as sugar, mills could use more of it for fermenting into ethanol. They were incentivised to do so by the Narendra Modi government paying more for ethanol made from B-heavy molasses and direct cane juice/syrup than from the normal C-heavy route. The result: Between 2013-14 and 2018-19, ethanol supplies to OMCs rose from 38 crore to 190 crore litres. So did the all-India average ethanol blending in petrol, from 1.6 per cent to 4.9 per cent.

The real change, however, happened when cereal grains began being used as ethanol feedstock. The process here involved first milling the grains into flour, extracting starch and breaking it down to simple sugars, followed by fermentation, distillation and dehydration to 99.9 per cent pure alcohol or ethanol suitable for blending with petrol.

Initially, the sugar mills themselves installed multi-feedstock distilleries to run on molasses and juice/syrup during the crushing period (November-April) and on grains in the off-season (May-October), when cane wasn't available. Over time, though, exclusively grain-based ethanol distilleries sprang up across the country. These used either maize or rice, which included surplus and old/broken/damaged grains sourced from the Food Corporation of India (FCI) as well

• ETHANOL SUPPLY BY FEEDSTOCK

(in Crore Litres)

Supply Year	C-Heavy Molasses	B-Heavy Molasses	Cane Juice	Maize	Surplus FCI Rice	Damaged Foodgrain	Total
2013-14	38	0	0	0	0	0	38
2014-15	67	0	0	0	0	0	67
2015-16	111	0	0	0	0	0	111
2016-17	67	0	0	0	0	0	67
2017-18	151	0	0	0	0	0	151
2018-19	146	33	1	0	0	10	190
2019-20	74	68	15	0	0	16	173
2020-21	39	183	39	0	2	39	302
2021-22	11	265	85	0	49	24	434
2022-23	6	235	128	32	74	32	507
2023-24	58	149	64	286	0	116	673
2024-25	17	139	165	498	140	80	1039
2025-26*	12	110	166	479	233	48	1048

*Allocated quantities. Supply years are Dec-Nov till 2021-22, Dec-Oct for 2022-23 and Nov-Oct from 2023-24.

as the open market.

In 2023-24, the total ethanol supply of 673 crore litres helped achieve an average 14.6 per cent blending level in petrol. Moreover, grain-based feedstock contributed 402 crore litres or 59.7 per cent. From being an alternative, off-season feedstock for distilleries attached to sugar mills, grains thus emerged as the mainstay of the EBP programme — a case of the tail wagging the dog.

For the current supply year ending October 2026, the OMCs have allocated about 1,048.3 crore litres of ethanol among various distilleries. Out of this quantity, just sufficient to meet a 20 per cent national blending target — E20 petrol — as much as 759.8 crore or 72.5 per cent is from grains and only the balance 288.5 crore (27.5 per cent) from sugarcane-based feedstock. Given the high sugar prices today and September-ending stocks projected at a 17-year-low, it is unlikely that the government will allow any ethanol production from direct cane juice or B-heavy molasses in the coming 2026-27 supply year.

Mills would be forced to recover the maximum possible sugar from cane juice, limiting the scope for making ethanol only from the non-crystallisable sucrose in C-heavy molasses. It will ensure no "diversion" of sugar through the B-heavy and cane juice routes. Such diver-

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sion amounted to 2.2 million tonnes (mt) in 2020-21 and 3.6 mt, 4.3 mt, 2.4 mt, 3.5 mt and 3 mt in the subsequent five sugar years (October-September). The 3 mt constituted roughly a tenth of India's estimated gross sugar output of 30.9 mt in 2025-26.

Simply put, we are heading into a scenario where the EBP programme will be almost exclusively fuelled by grains — the opposite of what it was till 2017-18. The accompanying table shows that even in 2021-22, the year when the average blending ratio crossed 10 per cent, sugarcane-based feedstock supplied over 83 per cent of ethanol.

From there, we have now moved to the tail not merely wagging, but taking full control over the dog. While E10 petrol has become E20, the government has already notified standards for E22, E25, E27 and E30 fuel blends. The push for it is basically coming from the distillers, who have set up an aggregate ethanol production capacity of nearly 2,000 crore litres, as against 421 crore in 2014 and the present annual offtake of 1,050 crore litres by OMCs.

It raises the question: Is there enough feedstock to fuel these ambitious targets? In 2026-27, it's not just sugarcane; even maize availability may present challenges, as El Niño gathers strength and with its effects expected to last through the first half of next year. That

leaves only rice — more so, the stocks lying in FCI godowns.

For 2025-26, the Modi government allocated 5.2 mt of surplus FCI rice to ethanol distilleries, which was raised to 7.2 mt in July. At 450-460 litres of ethanol production from every tonne of rice, the 7.2 mt would give 325-330 crore litres. With no sugar diversion and a possible tightening of maize supplies too, the only way to sustain even E20 that requires 1,050 crore litres of ethanol would be to earmark further quantities of FCI rice.

That would be very difficult to justify, not least because rice is a water-guzzling crop. Worse, FCI rice is being sold to ethanol distilleries at Rs 23.2 per kg, with the auction reserve price set even lower at Rs 21/kg for rice with 100 per cent broken grains. For comparison, the all-India modal or most-quoted retail market price for normal rice is Rs 40/kg and Rs 30/kg for broken rice, as per the latest Department of Consumer Affairs data.

Recall the EBP programme's original purpose: It was intended to help sugarcane growers by improving the liquidity of mills. The programme, likewise, provided an additional demand source for maize farmers whose produce was hitherto going mostly for poultry and livestock feed manufacturing. With ethanol, wholesale market prices of maize in India went up from Rs 13.8-17.8 per kg in 2021 to a Rs 22.1-24.5 range in 2024. There are two things the government could do.

First, discourage ethanol supplies from standalone grain-based distilleries, especially those significantly reliant on FCI rice. That party must end. Distillers should explore alternative, less water-consuming crops such as bajra and jowar. Their grains contain 58-62 per cent recoverable starch and can give 380-400 litres of ethanol from every tonne, the same as maize. Let millet farmers benefit from higher realisations like maize growers have.

Second, stop setting blending targets and trying to achieve them ahead of time. The EBP programme was successful even with 10-15 per cent blending and there's no need for posthaste. The government's chief economic adviser, V Anantha Nageswaran, in an article for this newspaper, has even suggested going back to E10. He's probably making a virtue out of today's reality.

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Problem With Ethanol Blending Isn't Sugar — It's Reliance on Grains

- Ethanol (ethyl alcohol) is a biofuel naturally produced by the fermentation of sugars by yeasts or via petrochemical processes.
- In India, it is primarily derived from sugarcane molasses, but increasingly from maize, surplus rice, and damaged foodgrains.
- Ethanol is classified into 1G, 2G, and 3G based on the type of feedstock used and the production technology.

First Generation (1G)	Edible biomass (Sugarcane, Corn, Wheat)	Competes with food security; technologically mature but ecologically taxing
Second Generation (2G)	Non-edible biomass (Agri-residue, Paddy Stubble, Bamboo)	Helps address crop burning (Punjab/Haryana); complex and costly conversion
Third Generation (3G)	Algae-based	High yield; no need for arable land or freshwater; still in R&D stage in India
Fourth Generation (4G)	Fourth Generation (4G) Genetically modified crops/organisms	Designed for higher carbon capture; largely theoretical at present

Problem With Ethanol Blending Isn't Sugar — It's Reliance on Grains

- **Blending:** It involves mixing ethanol with motor fuel (petrol) to reduce the consumption of imported crude oil and curb carbon emissions.
E10 and E20 petrol blended with 10%, 20% ethanol respectively, with higher blends reducing crude oil use and emissions.
- **The Ethanol Blended Petrol (EBP) Programme** aimed at mixing ethanol with standard motor petrol. Operating under the ambit of the National Biofuel Policy (NBP) 2018, the program is driven by the Ministry of Petroleum and Natural Gas.
- Its primary goals are to reduce India's heavy reliance on imported crude oil, cut greenhouse gas emissions, and boost farmers' incomes by creating a lucrative alternate market for agricultural produce.
- The program achieved a monumental milestone when India successfully hit its 20% ethanol blending target (E20) in 2025, a full five years ahead of its original 2030 deadline.

How Social Media Hooks Children – the Meta Settlement

How social media hooks children

Endless feeds, recommendation algorithms, notifications and 'likes': Social media design can severely affect mental health of young users

• IN META SETTLEMENT, A BLUEPRINT FOR INDIA?

AS CONCERNS around young users' exposure to social media increasingly enter policy discussions in India, Indian regulators are closely studying Meta's recent settlement with US states over addictive design.

THE SETTLEMENT is being examined at a time when the Centre is weighing a range of possible interventions around children's access to social media, including age-based restrictions, limits on the hours they can use such platforms and stronger parental consent requirements. These discussions are still at an early stage.

BUT THE US case could offer Indian policymakers a blueprint – instead of relying only on platforms' voluntary safety features, regulators have secured enforceable restrictions on how one of the world's largest social media



Companies can design and operate its services for minors.

THE SETTLEMENT could consequently feed into the wider debate in India on how much responsibility should rest with platforms themselves – rather than parents or children (who has effectively been tech companies' patch job) – for limiting potentially

harmful or compulsive use.

THE BROAD issue is not only in the payout, which is up to \$171 billion, but also in the product-level changes: usage caps, overnight restrictions, stronger age assurance and parental controls. Several of these ideas are also being discussed in India.

—SOURAVRENDRA BARIK



RINKU GHOSH

META, THE owner of Facebook and Instagram, has agreed to pay up to \$171 billion in penalties and make major changes to its products. From time limits and non-personalised feeds to parental controls and features to reduce constant engagement (see accompanying story).

Under the landmark settlement in the US, it will have to make a range of changes to its products. From time limits and non-personalised feeds to parental controls and features to reduce constant engagement (see accompanying story).

These changes will only be applicable to under-18s in US states and territories that participated in the case. However, it comes at a time when calls for banning social media use for children are gaining traction across the world, with the precedent set by a landmark Australian legislation that went into effect last year. India, too, is exploring graded, age-based restrictions.

What is interesting about the settlement is how it forces the company to focus on the very design of its products. Dr. Johannes Thürl, Associate Professor, Department of Mental Health, Johns Hopkins Bloomberg School of Public Health, said the development reflects a growing recognition that platform design can influence mental health outcomes, particularly for young users.

"I think this is a step in the right direction. The design requirements are more concrete than any other regulation I've seen in the US and include default night-time blocks, daily limits and tested age assurance," said Dr Thürl in an interview.

Excessive social media use Several studies show that mental health concerns among young people appear to be associated with excessive or problematic social media use.

"What we have seen in our research is that social media use directly is related to loneliness, depression and anxiety in young people. We have also found that reducing social media use or taking a break can improve mental health," says Dr Thürl.

Dr Thürl conducted a study involving 2,990 high-school students in Saudi Arabia and found that those with excessive social media use reported greater loneliness and mental distress than their peers. They also cited loss of control, continued use despite negative consequences,

interference with everyday life and negative effects on sleep. "Nearly a third of students screened positive for loneliness, and over 40% screened positive for depression or anxiety symptoms," he said.

A 2025 systematic review of 34 studies involving children and adolescents also found that off-task social media and smartphone use was generally associated with lower academic achievement, suggesting that frequent checking or scrolling can interfere with schoolwork and learning-focused

Addictive design

There is reasonable support to show that social media features designed to maximise user engagement can contribute to anxiety, depression, low self-esteem, sleep problems or other mental health concerns among adolescents. "Appearance-based social comparison, quantified feedback such as visible like counts, and overnight notifications have clear mechanisms of impact," said Dr Thürl.

The American Psychological Association specifically identifies features such as likes, recommended content, unrestricted time limits and endless scrolling as features that should be carefully considered when designing platforms for adolescents. A 2024 meta-analysis of 143 studies involving more than one million adolescents found that greater social media engagement was associated with higher levels of anxiety and depression.

Research has also found that the impact of social media depends partly on platform design, meaning that responsibility cannot be placed entirely on teenagers and their parents. The US Surgeon General has recommended that technology companies prioritise children's health and safety when designing their platforms and that governments introduce stronger safety standards.

Limiting social media use

There is experimental evidence supporting the idea that reducing social media and screen time can improve aspects of young people's mental health, particularly anxiety, depression, loneliness, sleep and emotional difficulties. A 2025 randomised controlled trial studied 220 young people with emotional distress. The group that reduced smartphone-based social media use to about one hour per day for three weeks experienced greater reductions in depression, anxiety and fear of missing out, as well as improvements in sleep.

A 2020 randomised controlled trial found that young people with symptoms of anxiety or depression, who reduced social media use by about 78 minutes per day, experienced a significant reduction in loneliness.

Why Meta's design changes matter more than settlement sum



ANAGHA JAYAKUMAR

META ON Wednesday agreed to pay more than \$171 billion to several US states over the next decade to settle claims of addictive platform design, misleading the public about associated harms and improperly collecting data from children under 13. As part of the settlement with 47 US states (as well as Washington DC and other territories) the company has promised to place strict limits on how teenagers use Facebook and Instagram.

The background

The settlement comes amid a much wider set of lawsuits over alleged harms caused by social-media platforms to children and teenagers. In March, a Los Angeles jury in *ROM v. Meta et al* held Meta and Google liable for \$6 million in damages after finding that features of their platforms contributed to a young user's mental-health harms.

Central to these cases is the design of these platforms themselves. Plaintiffs have pointed to endless feeds, recommendation algorithms, notifications and social metrics such as likes and views as features designed to keep young users engaged.

What Meta agreed to change

For the first five years of the settlement, Meta has agreed to impose a combined two-hour daily limit on Facebook and Instagram for users under 18, with direct messaging excluded from the limit. A verified parent can change the limit. Teen users will also be blocked from most parts of the platforms between midnight and 6 am, with most push notifications disabled between 10 pm and 7 am. Notifications will also be muted between 8 am and 1 pm on school weekdays. Direct messaging will remain available during the nighttime block.

The agreement also provides for a non-personalised feed option, hides likes and reactions by default, restricts cosmetic-procedure filters, and requires stronger age assurance, parental controls and safeguards around harmful content. These provisions address an issue that has already surfaced during earlier litigation: whether safety measures work when teenagers have to activate them themselves. At the *ROM* trial, a Meta document showed that only 11% of teen users used its existing daily-use limit.

The settlement makes some of these safeguards default settings, while others remain optional. Robbie Torney, Senior Director of AI Program at Common Sense Media, said that distinction could determine how widely they affect teenagers. "It's about the time and night mode limits to have a more effect simply because they're universal by default," Torney told *The Indian Express*. "The recommendation and engagement changes will likely reach fewer teens because a parent must go into the settings menu to enable them."

According to Carolina Rossini, Professor of Practice at UMass Amherst's School of Public Policy, the settlement is the first instance in the US where a major platform has been forced to change the architecture of its product, rather than its policies alone. "The dollar figure will get the headlines, but the consequential terms are those changes in design," she told *The Indian Express*. "A safety tool that requires you to find and turn on is fundamentally dif-

ferent from a constraint built into the product itself, and this agreement finally recognizes that distinction."

Accountability is key

Under the settlement, Meta must hire an independent auditor to assess compliance with the safeguards for five years. Whether these are effective in reducing harm remains to be seen. "Compliance and harm reduction are different questions, and we should refuse to let the first substitute for the second," Rossini said. "The real test is whether independent researchers get enough data access to measure outcomes: sleep, wellbeing, compulsive-use patterns, not feature-checklists."

Torney also cautioned against using screen time alone to judge whether the safeguards have worked. "Time spent on screens is a proxy, not the primary outcome that matters. The real question is what happens during the time teens are still on the app," he said. Torney pointed particularly to recommendation systems. "Maybe the question is how turning off algorithmic recommendations changes what teens see shown and re-engage with," he said.

The settlement also gives Meta a financial interest in seeing similar safeguards adopted elsewhere. About 30% of the \$171 billion is contingent on YouTube and TikTok adopting comparable protections and making matching payments. If Snapchat and YouTube all adopt comparable safeguards, stricter restrictions would follow, including a 60-minute daily limit on each platform.

Legal implications

However, the consent judgment explicitly states that the settlement is not an admission of wrongdoing by Meta and does not establish a standard of care or serve as precedent in any non-participating US state or international jurisdiction.

The settlement also leaves a major legal question unresolved. Before the agreement, Stanford Law professor Nora Freeman Engstrom said Meta's attempt to invoke Section 230 immunity against claims targeting platform design could be the settings menu to enable them. While an outcome in the states' trial would not have formally bound the roughly 200 other cases in the multi-state litigation, it could have shaped settlement negotiations and litigation strategies.

But Rossini cautioned against treating litigation as a substitute for regulation. "The settlement proves the design-accountability strategy works; it also proves we need to have legislative oversight to find and turn on is fundamentally dif-

How Social Media Hooks Children — the Meta Settlement

- **The Settlement**

- Meta agreed to pay up to \$17.1 billion in penalties and make major product changes over claims it endangered children with addictive design settling with 47 US States (plus DC and other territories) over misleading harm claims and improper data collection from under-13s.
- Design changes apply only to under-18s in participating US states, but Indian regulators are closely studying the settlement as a potential blueprint even as India separately weighs age-based restrictions and parental consent rules, still at an early stage.

- **Why the Design Changes Matter More Than the Money**

- For five years, Meta must impose a combined two-hour daily limit on Facebook/Instagram for under-18s (removable only by a verified parent); teens will be blocked from most platform access between midnight and 6 am, with notifications muted during school hours.
- The agreement also mandates a non-personalised feed option, hidden likes by default, restricted cosmetic filters, and stronger age-assurance addressing evidence that only 1.1% of teens used Meta's existing daily-limit tool when it was optional.

- **The Evidence Behind the Case**

- The case follows a wider set of lawsuits over social-media harms to children; in March, a Los Angeles jury held Meta and Google liable for \$6 million in damages after finding platform features contributed to a young user's mental-health harm.
- A 2025 review of 34 studies found off-task social media use linked to lower academic achievement; a study of 2,990 Saudi high-schoolers found excessive users reported greater loneliness and mental distress than peers.

India Seals Deal to Purchase Javelin Anti-Tank Missile System

India seals deal to purchase Javelin anti-tank missile system from US

Shows defence ties growing deeper, more capable, more consequential: Gor

Amrita Nayak Dutta
New Delhi, August 28

INDIA HAS finalised a deal to buy the Javelin anti-tank guided missile system from the United States. The Indian Army on Friday signed a Letter of Offer and Acceptance to buy the ordnance through the US Foreign Military Sales process.

Welcoming the decision, Washington said the “milestone” reflects the enduring partnership between the US and Indian armed forces, and opens the door for future collaboration between the defence industries.

“The landmark signature further strengthens the US-India Major Defence Partnership by demonstrating the depth of ongoing support to the Indian Army. It also opens discussions on mutually beneficial co-production and additional opportunities to strengthen the defence industrial bases of both countries,” the US Embassy said in a statement.

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What is the Javelin?



LOCKHEEDMARTIN.COM

The Javelin system is an advanced, ‘man-portable’, medium-range, fire-and-forget, anti-tank guided missile. It is designed for use against a wide array of targets, including armoured vehicles and fortified positions. Besides the US Army and US Marine Corps, it has several international customers.

“This agreement reflects a defence partnership growing deeper, more capable, and more consequential,” US Ambassador Sergio Gor said in a statement, adding that it “demonstrates depth of US support and creates new opportunities

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India Seals Deal to Purchase Javelin Anti-Tank Missile System

- **The FGM-148 Javelin is a man-portable, fire-and-forget ATGM, developed and produced by the Javelin Joint Venture of Lockheed Martin and RTX for the US military.**
- **It is primarily designed to destroy tanks and other armoured vehicles, but can also engage bunkers, fortified positions and buildings.**
- **Its infrared seeker locks onto the target before launch, eliminating the need for continuous guidance.**

Thank You